

Pharma-Khaato

Surgical Inventory & Credit RDBMS

Group Number: 6

Group Members:

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Project Overview

Pharma-Khaato is a database system built for surgical pharmacies. These shops sell medicines and surgical items and often give credit to regular customers using paper ledgers called *Khaatos*. This system replaces that paper process with a proper database that tracks stock, sales, and customer balances automatically.

Entities

- **Customers** — Stores the customer's CNIC, name, email, city, and registration date.
- **Employee** — Keeps a record of pharmacy staff members.
- **Products** — A list of all medicines and surgical items with price, stock quantity, minimum stock level, and chemical formula.
- **Supplier** — Contact details of companies that supply stock to the pharmacy.
- **Batch** — Each delivery of a product, with its own quantity, manufacture date, and expiry date.
- **Purchase** — A record of every sale, including total price, tax, discount, payment method, and which customer and employee were involved.
- **Sales_details** — Links each sale to the exact batch and quantity of product sold. Uses a composite primary key on purchase and batch.
- **Receipt** — Stores the date, time, and number of items for the receipt generated after each purchase.
- **Customer_Khatoo** — Tracks how much a customer has paid and how much they still owe, linked to both the customer and the purchase.

Key Relationships

- Each **Purchase** is made by one **Customer** and handled by one **Employee**.
- Each **Purchase** generates one **Receipt** and one **Customer_Khatoo** entry.
- A **Purchase** can include items from multiple **Batches** through **Sales_details**.
- Each **Batch** belongs to one **Product** and comes from one **Supplier**.
- **Customer_Khatoo** is directly linked to both the **Customer** and the **Purchase** it belongs to.
- **Employee** is linked to **Purchase** so the pharmacy can track who handled each sale.

Entity Relationship Diagram (ERD)

The diagram below shows the full database design for Pharma-Khaato using Crow's Foot notation.

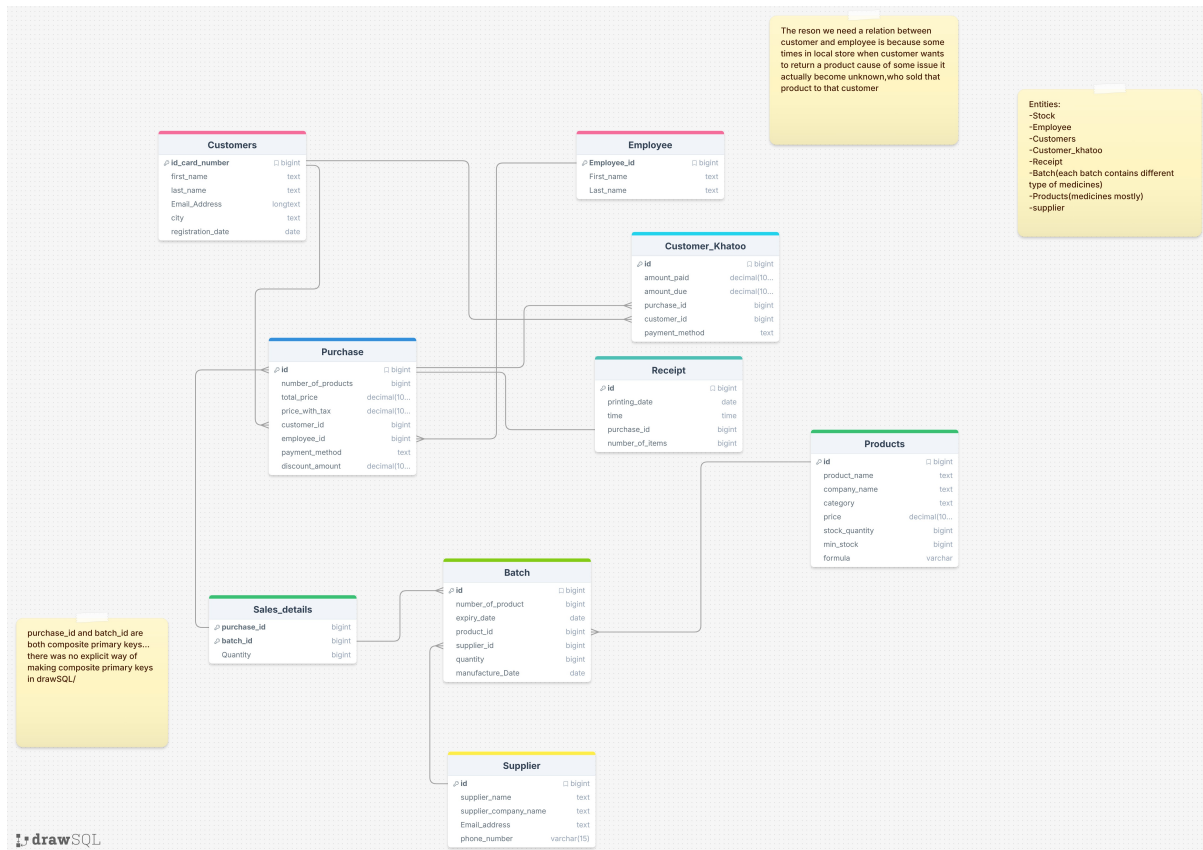


Figure 1: ERD of Pharma-Khaato (Crow's Foot Notation)